Chapter 1
The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Valid Arguments

An ***argument*** is a sequence of statements that end with a conclusion

Proofs in mathematics are ***valid arguments***

By ***valid***, we mean the conclusion must follow from the truth of the preceding statements (premises)

We use ***rules of inference*** to construct valid arguments

Valid Arguments in Propositional Logic

- Example:

Determine whether the following is a valid argument

"If you have a current password, then you can log onto the network."
"You have a current password."
"You can log onto the network."

Conclusion

Premises

Let p: "You have a current password"
q: "You can log onto the network"

- Argument form:

therefore

$$p \rightarrow q$$
$$p$$
$$\therefore q$$

Valid Arguments in Propositional Logic

- Example: $((p \rightarrow q) \wedge p) \rightarrow q$

tautology

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$((p \rightarrow q) \wedge p) \rightarrow q$
0	0	1	0	1
0	1	1	0	1
1	0	0	0	1
1	1	1	1	1

$p \rightarrow q$

p is a valid argument form

$\therefore q$

The sequence of propositions is a valid argument.

Valid Arguments in Propositional Logic

【Definition】

An **argument** in propositional logic is a sequence of propositions.

All but the final proposition in the argument are called **premises** and the final proposition is called the **conclusion**.

An argument is valid if the truth of all its premises implies that the conclusion is true.

An **argument form** in propositional logic is a sequence of compound propositions involving propositional variables.

An argument form is valid if no matter which particular propositions are substituted for the propositional variables in its premises, the conclusion is true if the premises are all true.

Valid Arguments in Propositional Logic

- The argument form with premises $p_1, p_2, \cdots, p_n$ and conclusion q

$$p_1$$
$$p_2$$
$$\vdots$$
$$p_n$$
$$\therefore q$$

therefore

is valid when $p_1 \wedge p_2 \wedge \cdots \wedge p_n \rightarrow q$ is a tautology

An argument in propositional logic is valid when its argument form is valid.

Rules of Inference for Propositional Logic

- Rules of inference are simple argument forms whose correctness we can establish with truth tables.

Rules of Inference for Propositional Logic

- Modus ponens (Latin for mode that affirms)

p	It is snowing today
$p \rightarrow q$	If it snows today, then we will go skiing
$\therefore q$	$\therefore$ We will go skiing

Rules of Inference for Propositional Logic

- Modus ponens (Latin for mode that affirms)

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

A valid argument can lead to an incorrect conclusion if one of its premises is wrong/false!

$$\begin{array}{l} \sqrt{2} > \dfrac{3}{2} \\ \sqrt{2} > \dfrac{3}{2} \rightarrow (\sqrt{2})^2 > \left(\dfrac{3}{2}\right)^2 \\ \hline \therefore 2 > \left(\dfrac{3}{2}\right)^2 \end{array}$$

Rules of Inference for Propositional Logic

- Modus tollens (Latin for mode that denies)

$\neg q$	We will not go skiing today
$p \rightarrow q$	If it snows today, then we will go skiing
$\therefore \neg p$	$\therefore$ It doesn't snow today

Rules of Inference for Propositional Logic

- Modus tollens (Latin for mode that denies)

$$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$$

Is this a tautology?

p	q	$\neg q$	$\boldsymbol{p \rightarrow q}$	$\boldsymbol{\neg p}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$
T	T	F	T	F	
T	F	T	F	F	
F	T	F	T	T	
F	F	T	T	T	

Rules of Inference for Propositional Logic

- Modus tollens (Latin for mode that denies)

$$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$$

p	q	$\neg q$	$p \rightarrow q$	$\neg p$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$
T	T	F	T	F	F T F
T	F	T	F	F	F T F
F	T	F	T	T	F T T
F	F	T	T	T	T T T

Rules of Inference for Propositional Logic

- Modus tollens (Latin for mode that denies)

1. $\neg q$
2. $p \rightarrow q$

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3. $\neg q \rightarrow \neg p$ Equivalence of Contrapositive, 2
4. $\neg p$ Modus ponens, 1,3

Rules of Inference for Propositional Logic

- **Hypothetical Syllogism**

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array} \qquad ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

- **Disjunctive syllogism**

$$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array} \qquad ((p \vee q) \wedge \neg p) \rightarrow q$$

- **Addition**

$$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array} \qquad p \rightarrow (p \vee q)$$

Rules of Inference for Propositional Logic

- **Simplification**

$$\frac{p \wedge q}{\therefore p} \qquad (p \wedge q) \rightarrow p$$

- **Conjunction**

$$\frac{p \quad q}{\therefore p \wedge q} \qquad ((p) \wedge (q)) \rightarrow p \wedge q$$

- **Resolution**

$$\frac{p \vee q \quad \neg p \vee r}{\therefore q \vee r} \qquad ((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$$

Using Rules of Inference to Build Arguments

- An argument is valid if
  - whenever all premises are true, the conclusion is also true
- To prove that an argument is valid:
  - Assume the premises are true
  - Use the rules of inference and logical equivalences to determine that the conclusion is true

Using Rules of Inference to Build Arguments

- **Example:** Show that the premises "It is not sunny this afternoon," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."

Solution:

Let p "It is sunn

s "We will tal

The argument fo

$\neg p$

$r \rightarrow p$

$\neg r \rightarrow s$

$s \rightarrow t$

$\therefore t$

Construct an argument by using rules of inference and logical equivalences

Step	Reason
1. $\neg p$	Premise
2. $r \rightarrow p$	Premise
3. $\neg r \rightarrow s$	Premise
4. $s \rightarrow t$	Premise
5. $\neg r$	Modus tollens, 1,2
6. s	Modus ponens, 3,5
7. t	Modus ponens, 4,6

Using Rules of Inference to Build Arguments

- **Example:** Show that $\neg w$ logically follows from the premises $(w \vee r) \rightarrow v, v \rightarrow (c \vee s), s \rightarrow u, \neg c \wedge \neg u$

Solution:

	Step	Reason
1.	$(w \vee r) \rightarrow v$	Premise
2.	$v \rightarrow (c \vee s)$	Premise
3.	$s \rightarrow u$	Premise
4.	$\neg c \wedge \neg u$	Premise
5.	$(w \vee r) \rightarrow (c \vee s)$	Hypothetical syllogism, 1,2
6.	$\neg u$	Simplification, 4
7.	$\neg s$	Modus tollens, 3, 6
8.	$\neg c$	Simplification, 4
9.	$\neg s \wedge \neg c$	Conjunction, 7,8
10.	$\neg (c \vee s)$	9 and De Morgan's law
11.	$\neg (w \vee r)$	Modus tollens, 5,10
12.	$\neg w \wedge \neg r$	10 and De Morgan's law
13.	$\neg w$	Simplification, 12

Using Rules of Inference to Build Arguments

- Note

If the conclusion is given in a form of $p \to q$, we can convert the argument

$$p_1 \wedge p_2 \wedge \cdots \wedge p_n \to (p \to q)$$

to

$$p_1 \wedge p_2 \wedge \cdots \wedge p_n \wedge p \to q$$

Because

$$(p_1 \wedge p_2 \wedge \cdots \wedge p_n \wedge p) \to q \equiv (p_1 \wedge p_2 \wedge \cdots \wedge p_n) \to (p \to q)$$

Using Rules of Inference to Build Arguments

- **Example**: Show that $r \rightarrow s$ logically follows from the premises $p \rightarrow (q \rightarrow s), \neg r \vee p, q$

Solution:

Step	Reason
1. $\neg r \vee p$	Premise
2. $p \rightarrow (q \rightarrow s)$	Premise
3. q	Premise
4. r	Additional premise
5. p	Disjunctive syllogism,1,4
6. $q \rightarrow s$	Modus ponens, 2, 5
7. s	Modus ponens, 3,6
8. $r \rightarrow s$	

Using Rules of Inference to Build Arguments

- Resolution rule

$$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore\ q \vee r \end{array}$$



Construct arguments in propositional logic using resolution as the only rule of inference

- Write the premises and the conclusion as clauses
- Use resolution rule

【Definition】 A clause is a disjunction of variables or negations of the these variables

- Example

  proposition $p \vee (q \wedge r)$ can be written as

  two clauses $p \vee q$ and $p \vee r$

Using Rules of Inference to Build Arguments

- Resolution rule
- Example: Show that the premises $(p \wedge q) \vee r$ and $r \rightarrow s$ imply the conclusion $p \vee s$.

Solution:

$(p \wedge q) \vee r \equiv (p \vee r) \wedge (q \vee r)$

$r \rightarrow s \equiv \neg r \vee s$

Premises: $p \vee r$, $q \vee r$, $\neg r \vee s$

$((p \vee r) \wedge (\neg r \vee s)) \rightarrow p \vee s$

Fallacies

- Examples

If you overslept, you'll be late.
You didn't oversleep.
Therefore: You aren't late.

$$p \rightarrow q$$
$$\neg p$$
$$\therefore \neg q$$

Fallacy of denying the hypothesis

If you are in China, you're in Asia.
You are in Asia.
Therefore: You are in China.

$$p \rightarrow q$$
$$q$$
$$\therefore p$$

Fallacy of affirming the conclusion

Rules of Inference for Quantified Statements

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TABLE 2 Rules of Inference for Quantified Statements.

Rule of Inference	*Name*
$\dfrac{\forall x\, P(x)}{\therefore\ P(c)}$	Universal instantiation
$\dfrac{P(c) \text{ for an arbitrary } c}{\therefore\ \forall x\, P(x)}$	Universal generalization
$\dfrac{\exists x\, P(x)}{\therefore\ P(c) \text{ for some element } c}$	Existential instantiation
$\dfrac{P(c) \text{ for some element } c}{\therefore\ \exists x\, P(x)}$	Existential generalization

Rules of Inference for Quantified Statements

- Examples:

1. Everyone in the discrete math class has taken a CS course. Marla is a student in the discrete class. Therefore, Marla has taken a CS course. Assume the domain consists of all people.

Solution:

Let $D(x)$: x is in the discrete math class
$C(x)$: x has taken a CS course

$$\begin{array}{l} \forall x(D(x) \rightarrow C(x)) \\ \underline{D(\text{Marla})} \\ \therefore\ C(\text{Marla}) \end{array}$$

1.	$\forall x(D(x) \rightarrow C(x))$	Premise
2.	D(Marla)	Premise
3.	D(Marla) $\rightarrow C(Marla)$	UI, 1
4.	C(Marla)	Modus ponens, 2,3

Rules of Inference for Quantified Statements

- Examples:

2. Each member of the committee is teacher and expert. Some members of the committee are young. Therefore, some members of the committee are young experts.

Solution:

Let $F(x)$ be "x is a mer
an expert," $H(x)$ be "x

$\forall x(F(x) \rightarrow G(x) \wedge H(x))$

$\exists x(F(x) \wedge R(x))$

$\therefore \exists x(F(x) \wedge G(x) \wedge R(x))$

Step	Reason
(1) $\exists x(F(x) \wedge R(x))$	Premises
(2) $\forall x(F(x) \rightarrow G(x) \wedge H(x))$	Premise
(3) $F(c) \wedge R(c)$	EI, 1
(4) $F(c) \rightarrow G(c) \wedge H(c)$	UI, 2
(5) $F(c)$	Simplification, 3
(6) $G(c) \wedge H(c)$	Modus ponens, 4,5
(7) $G(c)$	Simplification, 6
(8) $R(c)$	Simplification ,23
(9) $F(c) \wedge G(c) \wedge R(c)$	Conjunction,5,7,8
(10) $\exists x(F(x) \wedge G(x) \wedge R(x))$	EG,9

Combining Rules of Inference for Propositions and Quantified Statements

- Universal modus ponens

$$\forall x(P(x)\rightarrow Q(x))$$
$$P(a), \text{ where } a \text{ is a particular element in the domain}$$
$$\therefore Q(a)$$

- Example:

Assume that "For all positive integers n, if n is greater than 4,then n^2 is less than 2^n" is true. Use universal modus ponens to show that $100^2<2^{100}$.

Combining Rules of Inference for Propositions and Quantified Statements

- Universal modus tollens

$$\forall x(P(x) \rightarrow Q(x))$$

$\neg Q(a)$, where a is a particular element in the domain

$$\therefore \neg P(a)$$

Combining Rules of Inference for Propositions and Quantified Statements

- Example: Determine whether the following argument is valid.

(1) $\forall x \exists y G(x, y)$	Premise
(2) $\exists y G(a, y)$	UI using Step 1
(3) $G(a, c)$	EI using Step 2
(4) $\forall x G(x, c)$	UG using Step 3
(5) $\exists y \forall x G(x, y)$	EG using Step 4

Homework

Sec. 1.6 Due on Mar. 21 (Tuesday)

Ver. 7 & Ver. 8

12, 14(d), 18, 24, 29, 34(a)